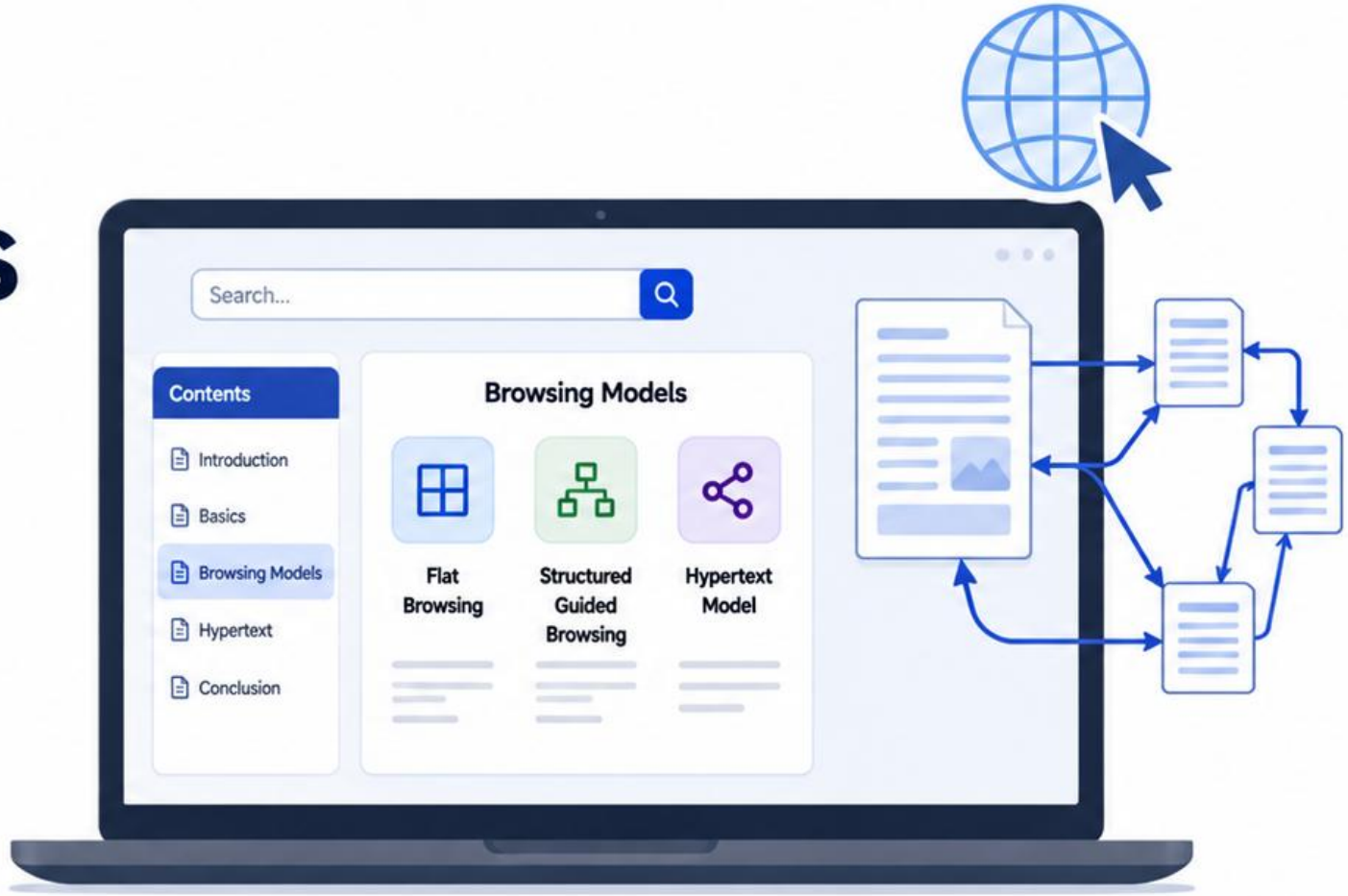
Week 5:
Browsing Models

Flat Browsing, Structured Guided Browsing, Hypertext Model

IR Focus:
Build browsing/navigation architecture for an FYP or software product

Advanced IR Focus:
Model browsing behaviour, user navigation paths, graph structure, and hypertext retrieval



Learning Outcomes

By the end of **Week 5**, students will be able to:

1. Explain browsing in Information Retrieval.
2. Differentiate flat, structured-guided, and hypertext browsing.
3. Design browsing/navigation architecture for a selected application.
4. Understand browsing behaviour in real-world IR systems.
5. Represent navigation using simple paths and graphs.
6. Connect browsing models with FYP modules, research papers, commercial products, and funded projects.

What Is Browsing in IR?

Browsing is an **interactive search activity** where the user moves through information based on feedback from the system.

In simple words:

Searching
is asking directly.

Browsing
is exploring step by step.

Example:

Searching: "Find lecture notes on Boolean Model."

Browsing:



Why Browsing Is Important

Browsing is important because users **do not always know** the exact query.

Users may:

- Explore available documents
- Move through categories
- Follow related links
- Check similar items
- Refine their need while navigating
- Discover useful information accidentally

A good IR system should support both **searching** and **browsing**.

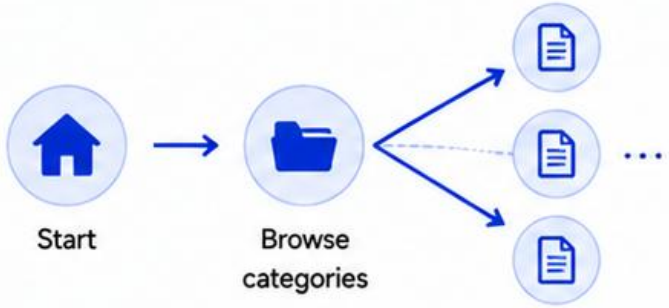


Types of Browsing Behaviour

There are **three common browsing behaviours**:

1 Exploratory Browsing

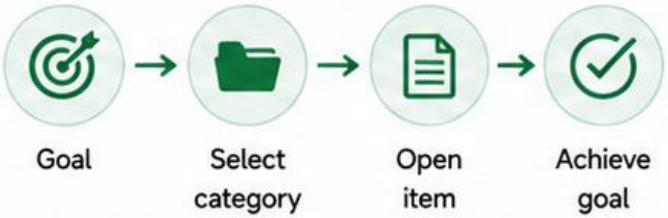
The user explores without a fixed target.



No fixed path – just exploration

2 Directed Browsing

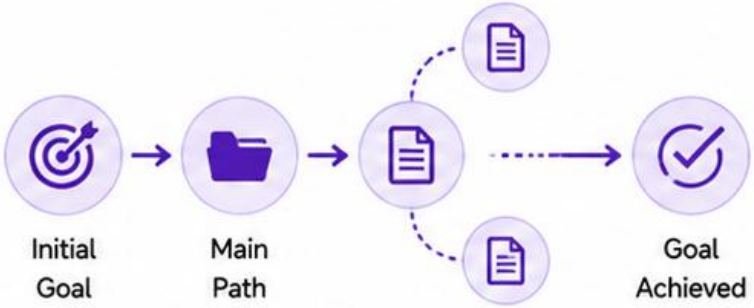
The user has a clear goal and follows a path.



Clear path – goal oriented

3 Opportunistic Browsing

The user starts with a goal but explores useful side information.



Mix of goal-directed and exploration

Real-World Examples

Google Search:
filters, images, videos,
news, time-based refinement.

Filters | Images | Videos | News | Time-based refinement

Amazon:
categories, filters, brands,
price range, customer ratings.

Categories | Filters | Brands | Price Range | Customer Ratings

YouTube:
thumbnails, related videos,
playlists, recommendations.

Thumbnails | Related Videos | Playlists | Recommendations

Academic Databases:
publication year,
authors, study type, citations,
related papers.

Publication Year | Authors | Study Type | Citations | Related Papers

FYP Portal Example: browse projects
by domain, supervisor, session,
technology, and funding potential.

Domain | Supervisor | Session | Technology | Funding Potential

Flat Browsing

Flat browsing means exploring documents **without a deep structure.**



A user may move document by document or read a single document from beginning to end.

Strength: Simple and easy.

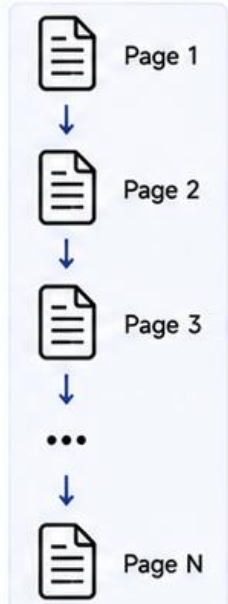
Weakness: User may lose context in large collections.

Example:

A student opens one PDF and scrolls from page 1 to the end to find useful content.



Scrolling from Start to End



Flat Browsing Computed Example

In flat browsing, the user reads documents one by one without deep navigation.
We can compute total browsing time to analyse user effort.

Suppose a user reads three documents:



- **D1** = 30 minutes — 30 min
- **D2** = 10 minutes — 10 min
- **D3** = 45 minutes — 45 min

Total browsing time:

Browsing Time = D1 + D2 + D3

Browsing Time = 30 + 10 + 45 = **85 minutes**

This helps analyse user effort in an IR system.

User → Reads D1 (30 min) → Reads D2 (10 min) → Reads D3 (45 min) → Total Time (85 min)

Why useful?

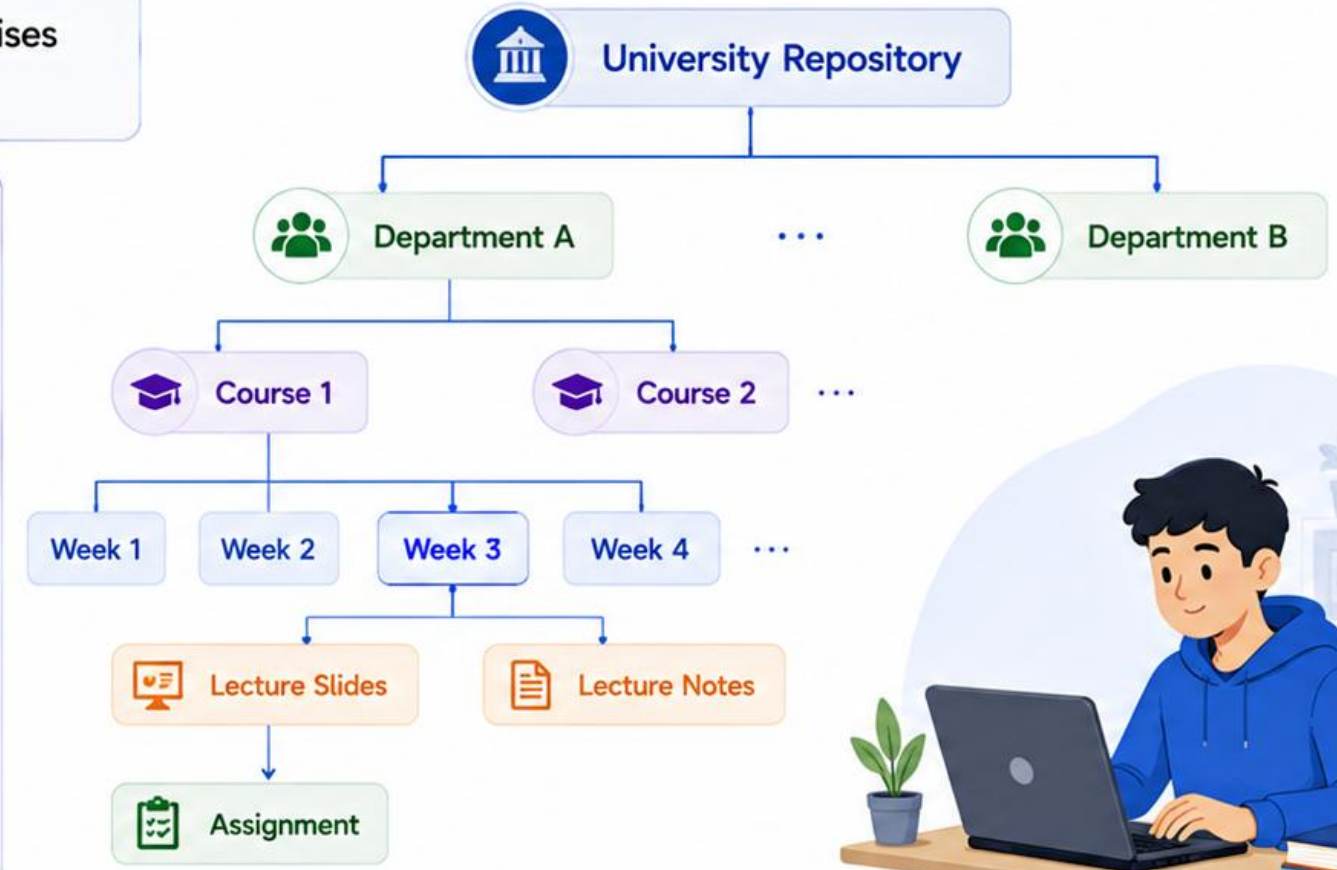
Helps compare browsing strategies, estimate user effort, and improve system design.

Structured Guided Browsing

Structured guided browsing organises documents in a hierarchy.

Example:

- University Repository
- ↓ Department
- ↓ Course
- ↓ Week
- ↓ Lecture Slides
- ↓ Assignment





This is useful when documents naturally have categories, folders, chapters, sections, or modules.

Structured Guided Browsing — Application View

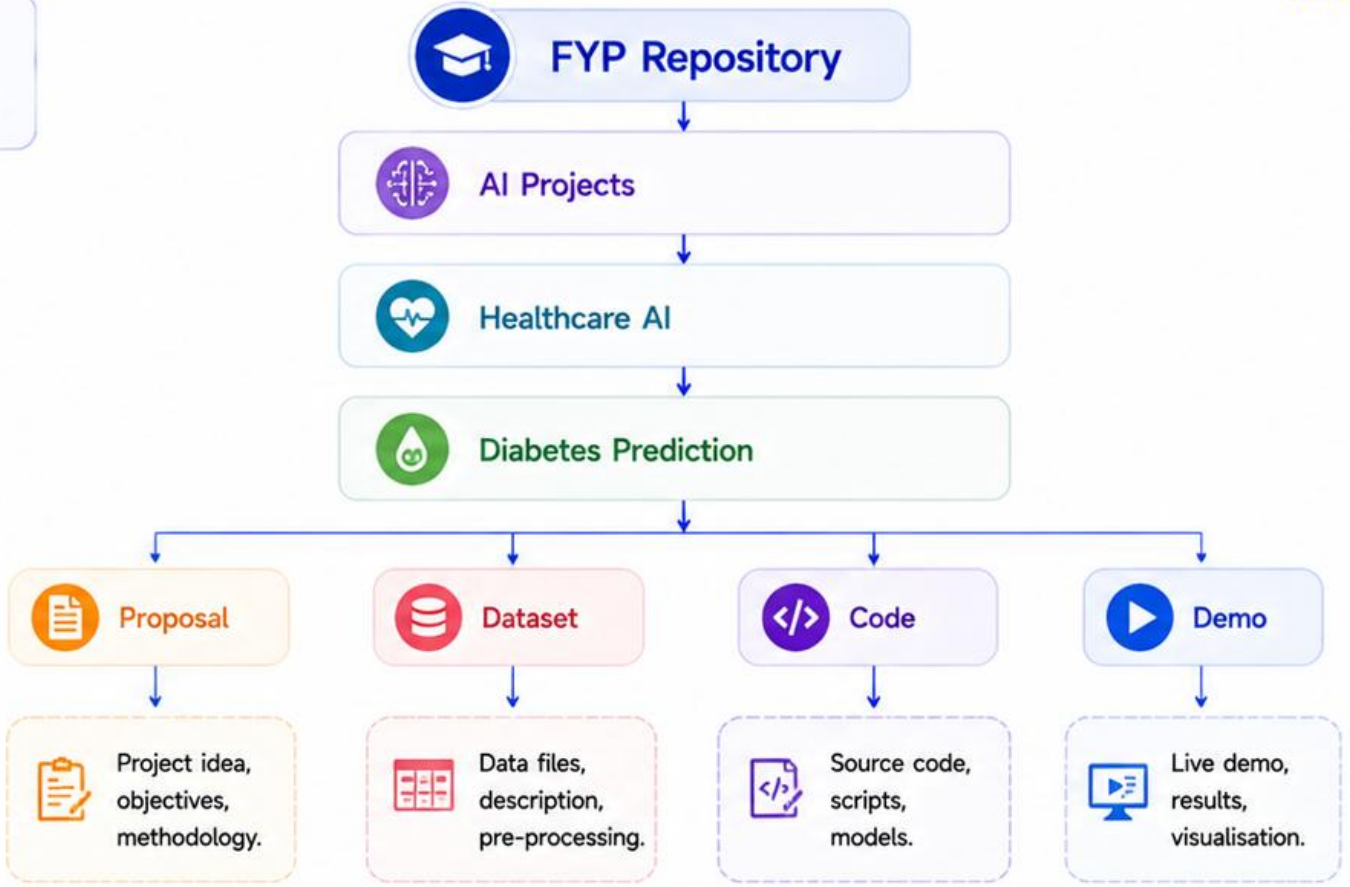
In an FYP system, documents are organised in a hierarchy for easy navigation.

FYP System Example

Structured guided browsing can be:

- FYP Repository
- AI Projects
- Healthcare AI
- Diabetes Prediction
- Proposal
- Dataset
- Code
- Demo





This makes the system easier to explore and improves user experience.

Structured Guided Browsing — Mathematical View

Structured guided browsing can be expressed mathematically.

A simple expression is:

$$\text{StructuredGuidedBrowsing}(D, H, UI) = N$$

Where:

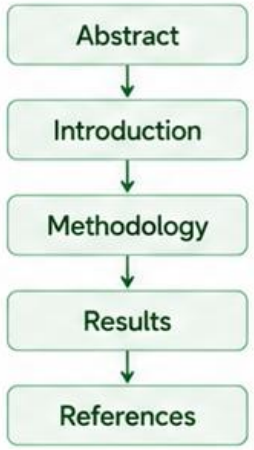
- **D** = Document or document collection
- **H** = Hierarchy or structure
- **UI** = User interface elements
- **N** = Navigation path followed by the user

Example:

D = Research Paper



H = Hierarchy / Structure



UI = User Interface Elements

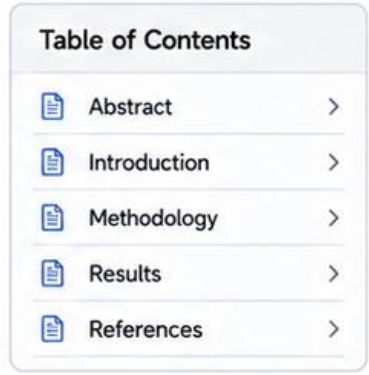
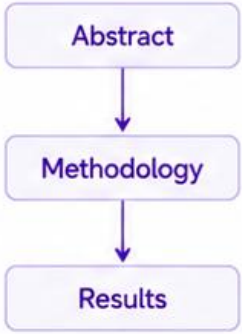


Table of contents and clickable headings

N = Navigation Path



N = Abstract → Methodology → Results

This mathematical view helps in **designing, analysing, and evaluating** structured browsing systems.

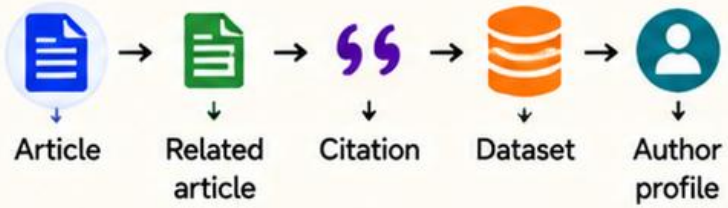
Hypertext Model

The Hypertext Model allows non-linear navigation through nodes and links.

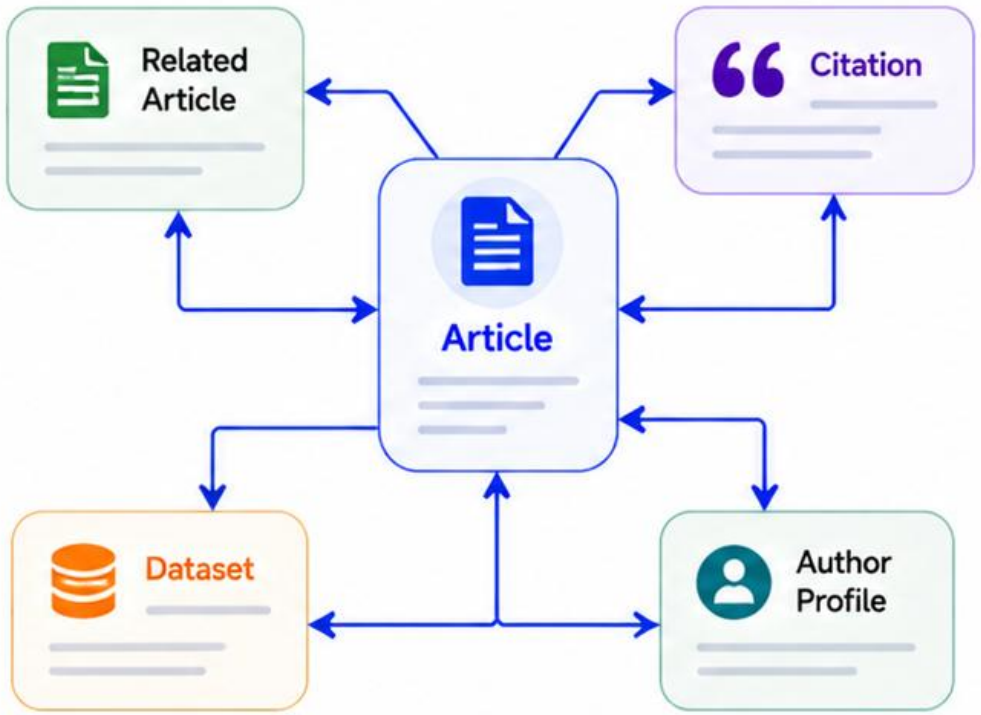
A user does not have to read from start to end.

The user can jump from one part to another through links.

Example:



How it Works



Key Points

- Non-linear navigation
- User-driven exploration
- Links connect related content
- Improves discovery and learning

The Hypertext Model mimics how we browse the web, making information access flexible and user-friendly.

IR Development Focus

For IR students, the target is to build a powerful **browsing / navigation module** in an Information Retrieval system.

Required Features

1 Category Menu



Organize documents in categories and subcategories.

2 Search + Browse Option



Allow users to search or browse through collections.

3 Breadcrumb Path



Show the path to help users understand their location.

4 Related Documents



Display documents that are related to the current one.

5 Recently Viewed Documents



Show a list of documents recently accessed.

6 Filters



Filter by type, date, author, topic, language, and more.

7 Navigation History



Let users go back and revisit previous pages.

8 Document Preview



Show a quick preview / snippet before opening the document.

9 Internal Links



Enable links within documents to improve exploration.

10 User-Friendly Interface



Clean, responsive and easy-to-use interface for all users.

Advanced IR Research / Product Focus

For Advanced IR students, browsing should be **treated as a model**.



Possible Research/Product Directions

1 User Browsing Behaviour Analysis

Analyze how users browse, what they click, how long they stay, and what patterns they follow.

2 Navigation Path Prediction

Predict the next document a user will visit based on current navigation context and history.

3 Graph-based Document Navigation

Model documents as a graph and use graph algorithms to support intelligent navigation.

4 Hypertext-based Academic Paper Retrieval

Use hyperlinks, citations and sections to retrieve and navigate academic papers effectively.

5 Personalised Browsing Recommendations

Recommend documents, links or sections based on user interests, profile and past browsing behaviour.

6 Search + Browse Hybrid Retrieval Model

Combine keyword/semantic search with browsing context to deliver more relevant results and paths.

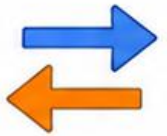
7 Fundable Digital Library or FYP Repository Product

Build a complete digital library or FYP repository system with advanced browsing features for real-world impact.

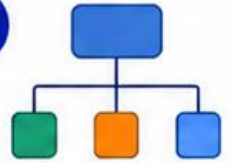
PART 8 – LEARNING CHECK

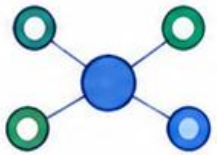
Quiz

1.  What is browsing in IR?

2.  Difference between search and browsing?

3.  What is flat browsing?

4.  Why is structured guided browsing useful?

5.  What are nodes and links in the Hypertext Model?



Discussion | Discuss your answers with your classmates before sharing them with the instructor.

Case Study Assignment

For IR Students:

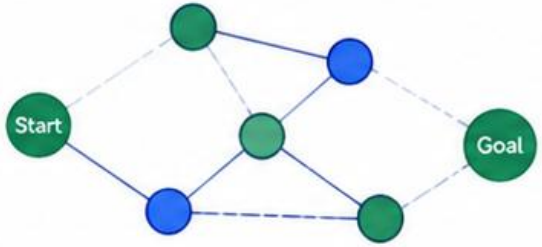
Design and implement browsing/navigation architecture for a selected FYP application.

Minimum Requirements

1. Home page
2. Category hierarchy
3. Document list
4. Search + browse option
5. Breadcrumb navigation
6. Related document links
7. Document preview

For Advanced IR Students:

Prepare a mathematical/research note on browsing as a navigation model using paths, nodes, links, and graph representation.



- Node
- Link / Edge
- Path

Include in your note:

- Definitions and assumptions
- Graph representation (G = (V, E))
- Example browsing paths
- Possible algorithms / metrics for navigation

PART 8 – SUMMARY

Summary

Week 5 covered:

1. Browsing behaviour
2. Flat browsing
3. Structured guided browsing
4. Hypertext Model
5. Navigation path
6. Nodes and links
7. FYP browsing architecture
8. Research/product direction for browsing models

These concepts and architectures form the foundation for intelligent navigation and advanced information retrieval systems.